

Vulnerability Assessment Report

February 25th, 2026

System Description

The system consists of a remote database server that stores customer information for the company's e-commerce operations. The server is accessible over the internet to support employees working remotely from multiple global locations. Employees regularly query the database to identify and manage potential customers. The server operates with public network accessibility and allows external connections without sufficient access restrictions. The database environment supports core business functions and interacts with employee endpoints through remote network connections. Existing security measures include basic connectivity controls and encrypted data transmission.

Scope

The scope of this vulnerability assessment focuses on the public accessibility of the remote database server and its associated logical access controls. The assessment evaluates risks related to unauthorized access, data exposure, and potential impact on business operations. The analysis considers the confidentiality, integrity, and availability of customer information stored on the server. The assessment follows general risk evaluation principles to identify threats associated with external exposure and to support recommendations for securing remote database access.

Purpose

Consider the following questions to help you write:

- *How is the database server valuable to the business?*
- *Why is it important for the business to secure the data on the server?*
- *How might the server impact the business if it were disabled?*

The purpose of this assessment is to evaluate the security risks created by the public accessibility of the company's remote database server, which supports core e-commerce operations by storing and providing customer information. Because the database enables employees to identify prospects and perform daily tasks, protecting the confidentiality, integrity, and availability of this data is essential to business continuity and customer trust. This assessment identifies how unauthorized access or service disruption could impact operations and supports informed risk management decisions by leadership.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
<i>E.g. Competitor</i>	<i>Obtain sensitive information via exfiltration</i>	1	3	3
<i>Hacker</i>	<i>Conduct “Man-in-the-middle” attack</i>	3	3	9
<i>Employee</i>	<i>Alter-delete critical information</i>	1	3	3

Approach

This assessment evaluates high-impact threats to the public database server. External competitors could exfiltrate sensitive information, undermining market position and customer trust. Malicious hackers may execute man-in-the-middle attacks, intercepting credentials and data in transit, representing the highest risk. Insider employees could alter or delete critical records, affecting integrity and availability. Together, these scenarios demonstrate significant exposure across confidentiality, integrity, and availability, justifying urgent remediation and strengthened access controls, monitoring, and encryption. Immediate action is required to reduce organizational risk.

Remediation Strategy

Remediation should start by removing public access to the database and allowing entry only to authorized staff. Employees should have access only to the information they truly need to do their jobs. Strong passwords and an extra verification step should be required to log in. Data should be protected so it cannot be easily read if intercepted. The company should monitor activity to detect unusual behavior early. Regular updates, backups, and simple security training will help prevent mistakes, reduce risk, and ensure the business can quickly recover if an incident occurs.